

Lecture 1: Introduction to Longitudinal Data Analysis

January 23nd, 2025

Class Outline

- Introductions
 - Myself
 - Class CA's
- Class format
- Protocol for if you feel sick
 - Syllabus review (lets all get on the same page)
- What is longitudinal data?
 - Why do we use it?
 - What types of questions can we answer?
- Data sources (oh the many data sources)

Introductions

Class format

- **In person.**
- If you are sick, email CAs and myself beforehand, review slides, seek out help from CAs if you have questions.
 - IF YOU FEEL SICK, STAY HOME!
- Your CA will take attendance at every in person session starting after the drop/add date.
 - Part of your participation grade
 - You will not be penalized for not coming to class for being sick
 - Everyone gets 1 missed classes, no questions asked

Syllabus Review

- See screen share.....

Why Do We Use Longitudinal Data?

- We want to understand how something changes over time
 - Studies in which individuals are measured repeatedly through time.
- Need metrics to do this
 - Months, semesters, days, years...etc.
- We are interested in knowing the functional form of a phenomena

Features of Longitudinal Data

- Key feature of longitudinal studies is that measurements of the same individuals are taken repeatedly through time.
 - Longitudinal studies allow direct study of change over time.
- Objective: characterize the change in response over time and the factors that influence change.
- With repeated measures on individuals, we can capture within-individual variations.

Example 1: Treatment of Lead-Exposed Children Trial

- Exposure to lead during infancy is associated with substantial deficits in tests of cognitive ability
- Traditional treatment of children with high lead levels usually requires injections and hospitalization
- A new agent, Succimer, can be given orally
- Randomized trial examining changes in blood lead level during course of treatment
- 100 children randomized to placebo or Succimer
- Measures of blood lead level at baseline, 1, 4 and 6 weeks

Table 1: Blood lead levels ($\mu\text{g}/\text{dL}$) at baseline, week 1, week 4, and week 6 for 8 randomly selected children.

ID	Group ^a	Baseline	Week 1	Week 4	Week 6
046	P	30.8	26.9	25.8	23.8
149	A	26.5	14.8	19.5	21.0
096	A	25.8	23.0	19.1	23.2
064	P	24.7	24.5	22.0	22.5
050	A	20.4	2.8	3.2	9.4
210	A	20.4	5.4	4.5	11.9
082	P	28.6	20.8	19.2	18.4
121	P	33.7	31.6	28.5	25.1

^a **P = Placebo; A = Succimer.**

Table 2: Mean blood lead levels (and standard deviation) at baseline, week 1, week 4, and week 6.

Group	Baseline	Week 1	Week 4	Week 6
Succimer	26.5 (5.0)	13.5 (7.7)	15.5 (7.8)	20.8 (9.2)
Placebo	26.3 (5.0)	24.7 (5.5)	24.1 (5.7)	23.2 (6.2)

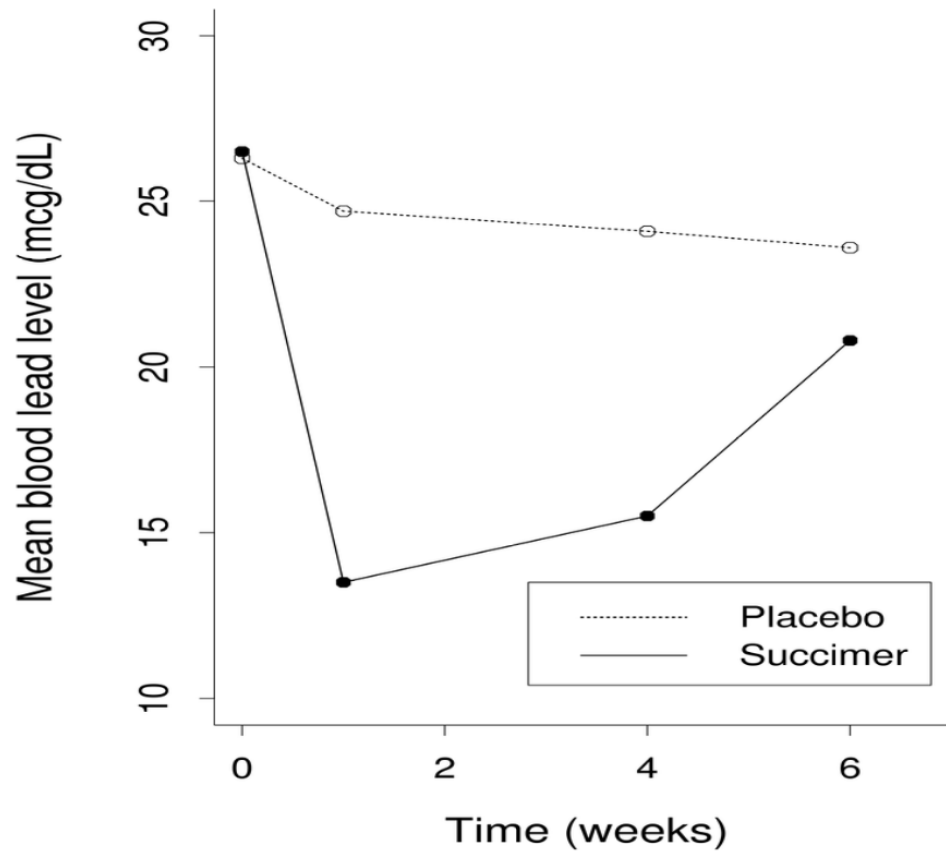


Figure 1: Plot of mean blood lead levels at baseline, week 1, week 4, and week 6 in the succimer and placebo groups.

Example 2: Six Cities Study of Air Pollution and Health

- Longitudinal study designed to characterize lung function growth in children and adolescents.
- Most children were enrolled between the ages of six and seven and measurements were obtained annually until graduation from high school.
- Focus on a randomly selected subset of the 300 female participants living in Topeka, Kansas.
- Response variable: Volume of air exhaled in the first second of spirometry manoeuvre, FEV1.

Table 3: Data on age, height, and FEV_1 for a randomly selected girl from the Topeka data set.

Subject ID	Age	Height	Time	FEV_1
159	6.58	1.13	0.00	1.36
159	7.65	1.19	1.06	1.42
159	12.74	1.49	6.15	2.13
159	13.77	1.53	7.19	2.38
159	14.69	1.55	8.11	2.85
159	15.82	1.56	9.23	3.17
159	16.67	1.57	10.08	2.52
159	17.63	1.57	11.04	3.11

Note: Time represents time since entry to study.

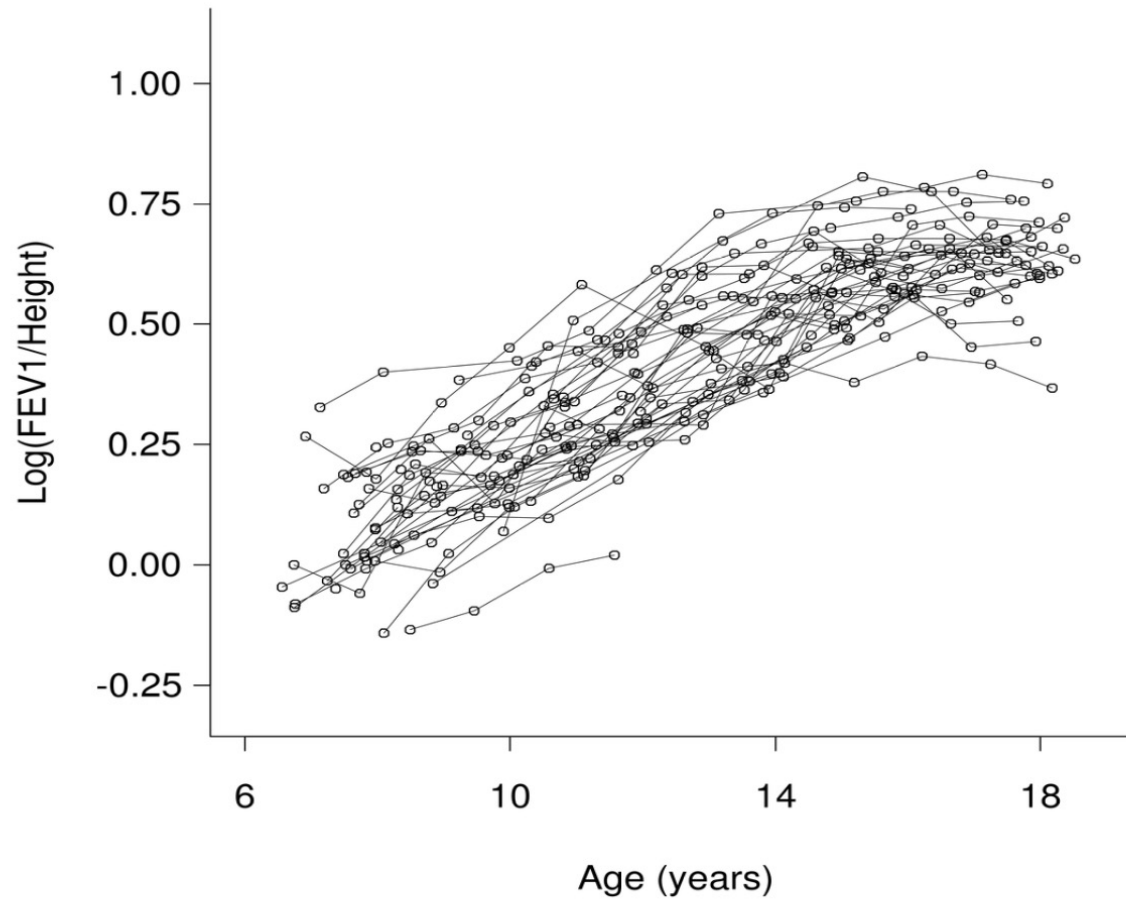


Figure 2: Timeplot of $\log(\text{FEV}_1/\text{height})$ versus age for 50 randomly selected girls from the Topeka data set.

Example 3: Clinical Trial of Anti-Epileptic Drug Progabide

- Randomized, placebo-controlled study of treatment of epileptic seizures with progabide.
- Patients were randomized to treatment with progabide, or to placebo in addition to standard therapy.
- Outcome variable: Count of number of seizures.
- Measurement schedule: Baseline measurement during 8 weeks prior to randomization. Four measurements during consecutive two-week intervals.
- Sample size: 28 epileptics on placebo; 31 epileptics on progabide.

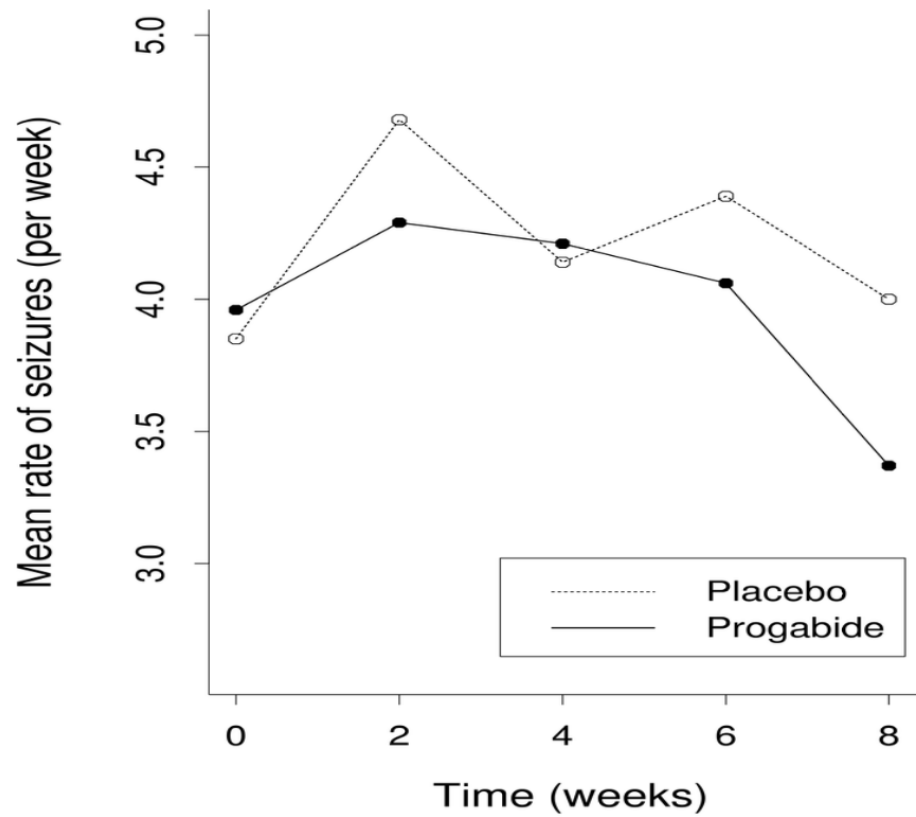


Figure 5: Mean rate of seizures (per week) at baseline, week 2, week 4, week 6, and week 8 in the progabide and placebo groups.

More Examples

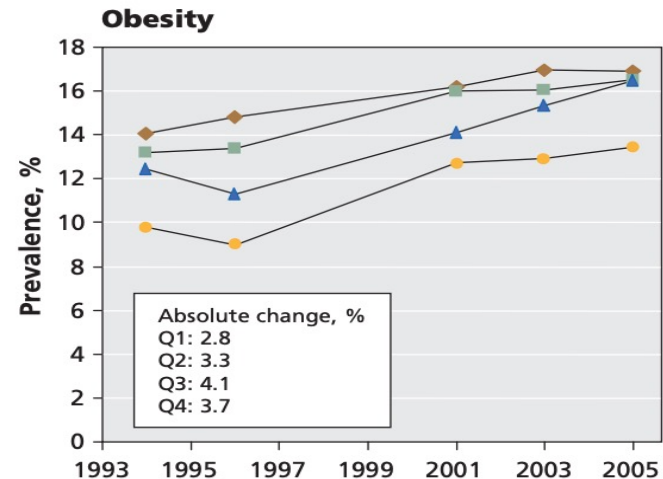
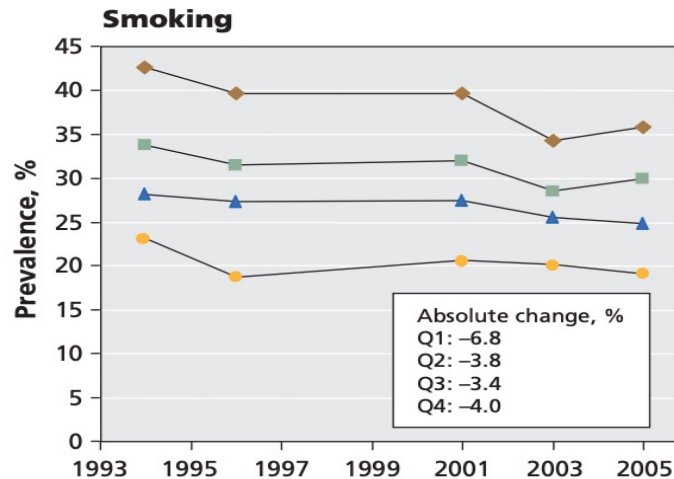
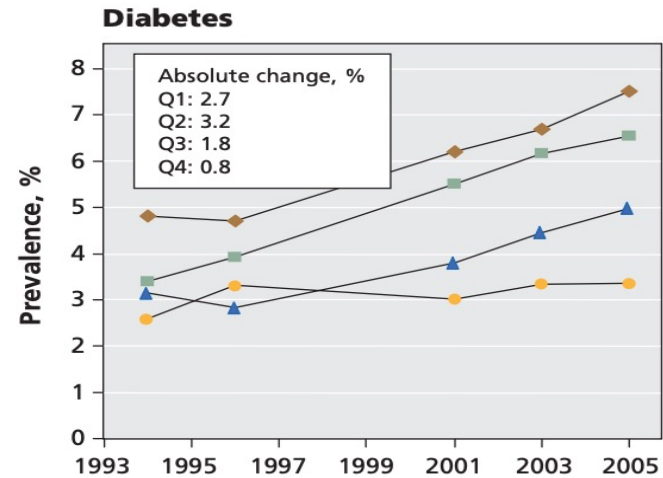
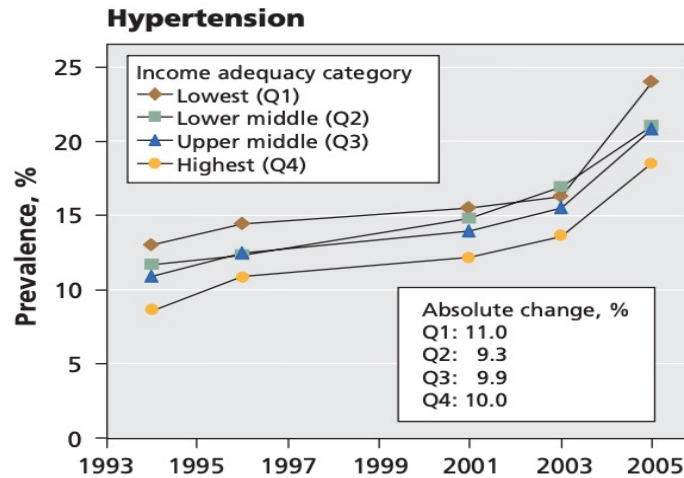
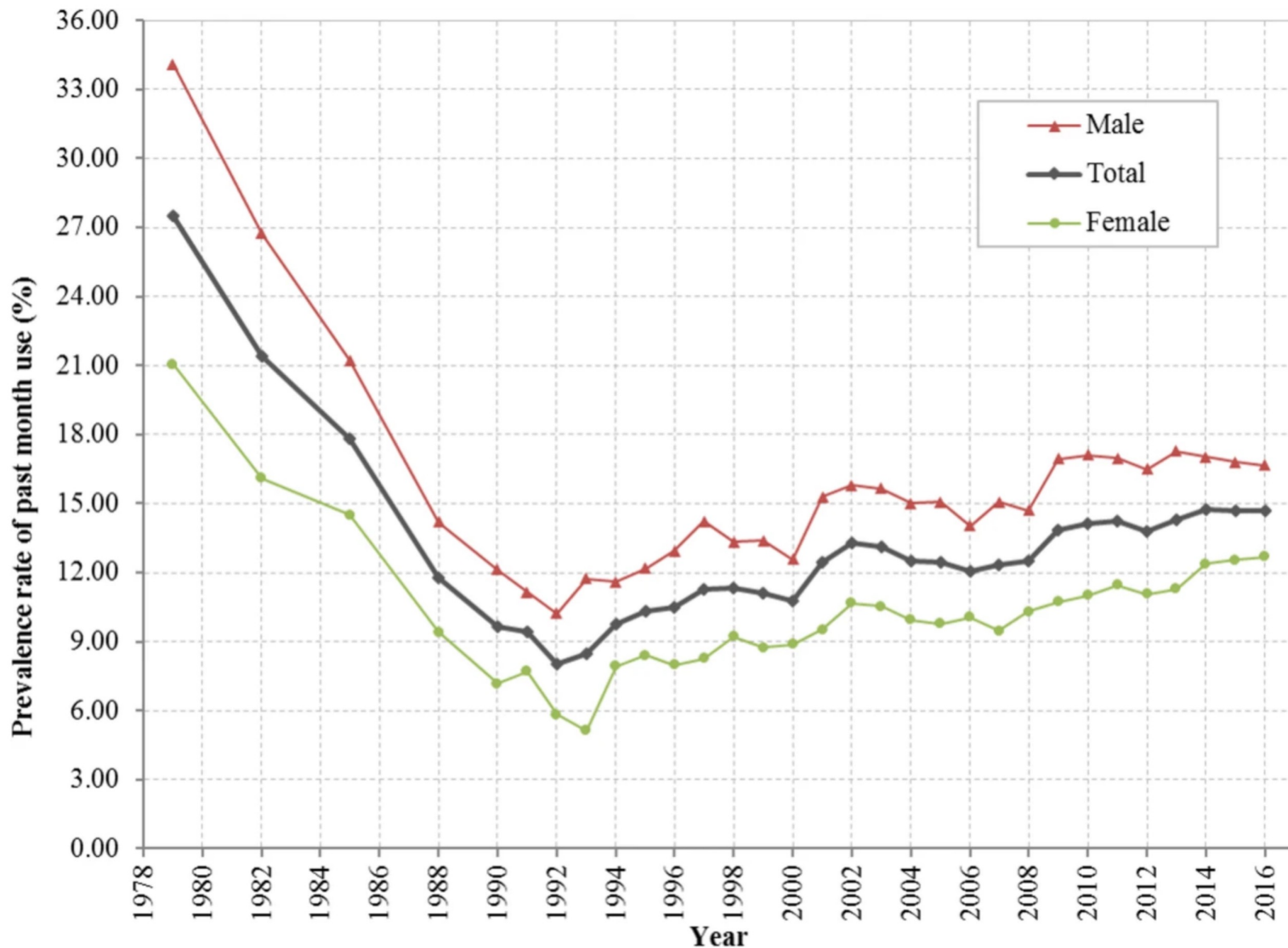


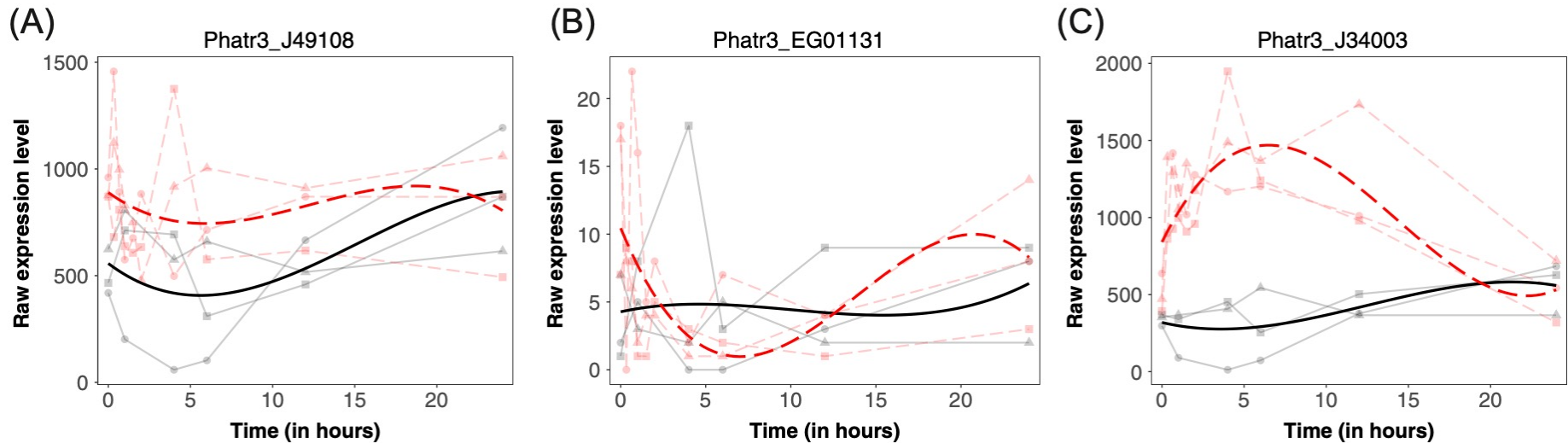
Figure 2: Age- and sex-adjusted prevalence of hypertension, diabetes, smoking and obesity from 1994 to 2005, stratified by income adequacy category.

Fig. 1

From: [Marijuana legalization and historical trends in marijuana use among US residents aged 12–25: results from the 1979–2016 National Survey on drug use and health](#)



Prevalence rate (%) of current marijuana use among US residents 12 to 25 years of age during 1979–2016, overall and stratified by gender. Derived from data from the 1979–2016 National Survey on Drug Use and Health (NSDUH)



Three types of Differential Expression (DE) profiles: (A) PDE with no time-by-treatment, (B) NPDE with only time-by-treatment, and (C) NPDE with both treatment and time-by-treatment across time using the bulk RNA-seq data for P.t. from a longitudinal experiment on the effect of light intensity on the algae physiology and molecular mechanism.

Two light conditions are examined in this experiment, low and high light levels. Red and black curves represent the group with low and high light levels, respectively.

Types of Longitudinal Data

1. Panel Studies

- Follow individual over time who start at the same time and data collected at **uniform** intervals.
 - Panel Study of Income Dynamics

2. Retrospective Studies

- Look back to understand what exposure led to the outcome.
- Usually used in a **preliminary** manner
- **Used when the outcome is rare**
 - Developmental causes of a rare form of cancer among late adolescents

3. Cohort (Clustered) Studies

- Collecting data from a group of people with a shared characteristic
- Usually interested in establishing **a causal association** so an intervention can be randomly given to one group vs. another

- Individuals/Subjects: Participants in a longitudinal study are referred to as individuals or subjects.
- Occasions: In a longitudinal study individuals are measured repeatedly at different occasions or times.
 - The number of repeated observations, and their timing, can vary widely from one longitudinal study to another.
 - When number and timing of the repeated measurements are the same for all individuals, study design is said to be “balanced” over time.
- Designs can be balanced, although studies may have incompleteness in data collection.

Examples of Longitudinal Questions

1. Does the keto diet lead to long term weight loss?
2. Does going to a low-income elementary school increase the risk of substance abuse behavior in early adulthood?
3. Does fluoride fortification during adolescence lead to a reduction in cavities across the life course?
4. Do states with a larger population of deer have more long term fruit scarcity?

Advantages of Longitudinal Data

1. Longitudinal studies help you discover variable patterns over time, leading to more precise causal relationships and research outcomes.
2. When researching developmental trends, longitudinal studies allow you to discover changes across lifespans and arrive at valid research outcomes.
3. They are highly flexible, which means the researcher can adjust the study's focus while it is ongoing.
4. Unlike other research methods, longitudinal studies collect unique, long-term data and highlight relationships that cannot be discovered in a short-term investigation.
5. You can collect additional data to study unexpected findings at any point in your systematic investigation.

Advantages of Longitudinal Data

- Ability to control heterogeneity
 - Can address dependence across units and levels of those units
 - If you are trying to understand weather changes across 5 states in the US over a 10 year period it is probably the case that the weather in state 1 is related to the weather in state 2.
- Ability to measure effects that are not detectable in cross-sectional data
 - The effect of time; The history effect
- More information, variability, more efficiency
- In longitudinal studies the outcome variable can be:
 - continuous (e.g., blood lead levels)
 - binary (e.g., presence/absence of onycholysis)
 - count (e.g., number of gene expression level)

Draw backs of longitudinal data

1. It's difficult to predict the results of longitudinal studies because of the extended time frame. Also, it may take several years before the data begins to produce [observable patterns](#) or relationships that can be monitored.
2. It costs lots of money to sustain a research effort for years. You'll keep incurring costs every year compared to other forms of research that can be completed in a smaller fraction of the time.
 - Subjects may be measured at different occasions (e.g., due to mistimed measurements).
3. Longitudinal studies require a large sample size which might be challenging to achieve. Without this, the entire investigation will have little or no impact.
4. Longitudinal studies often experience panel attrition. This happens when some members of the research sample are unable to complete the study due to several reasons like changes in contact details, refusal, incapacity, and even death.
 - The data set can be incomplete (missing data/dropout).
5. Statistical dependence (e.g. 1 individual with multiple observations)

Data Sources

- The National Longitudinal Study of Adolescent to Adult Health (Add Health)

<https://addhealth.cpc.unc.edu/>

- Applied Longitudinal Analysis Datasets

<https://content.sph.harvard.edu/fitzmaur/ala2e/>

- Flint Adolescent Study

<https://fas.sph.umich.edu/>

- ICPSR (this is a data repository)

<https://www.icpsr.umich.edu/web/pages/>

All of Us Research Program (NIH)

- **Diversity at the scale of 1 million people:** demographically, geographically, medically, and especially those underrepresented in biomedical research
- **Diversity of data types collected longitudinally:** clinical, environmental, genetic, behavioral, socioeconomic
- **National, open resource for all:** open to the public and all researchers, open source software & tools

<https://databrowser.researchallofus.org/>

Example: Race and Discrimination In the Time of COVID-19

Outcome variable

Major Experiences of Discrimination

Independent variable

Race/ethnicity

Research Question

Does race/ethnicity predict trajectories of discrimination during the COVID-19 epidemic?

Hypothesis

Asian and Black Americans will have have higher mean levels of discrimination at baseline compared to White Americans

Asian and Black Americans will have positive linear trajectories of discrimination over the 5 waves of data compared to White Americans.

Distribution of Race/ethnicity

From monthly survey 2020 May to September

Total sample size: 66220

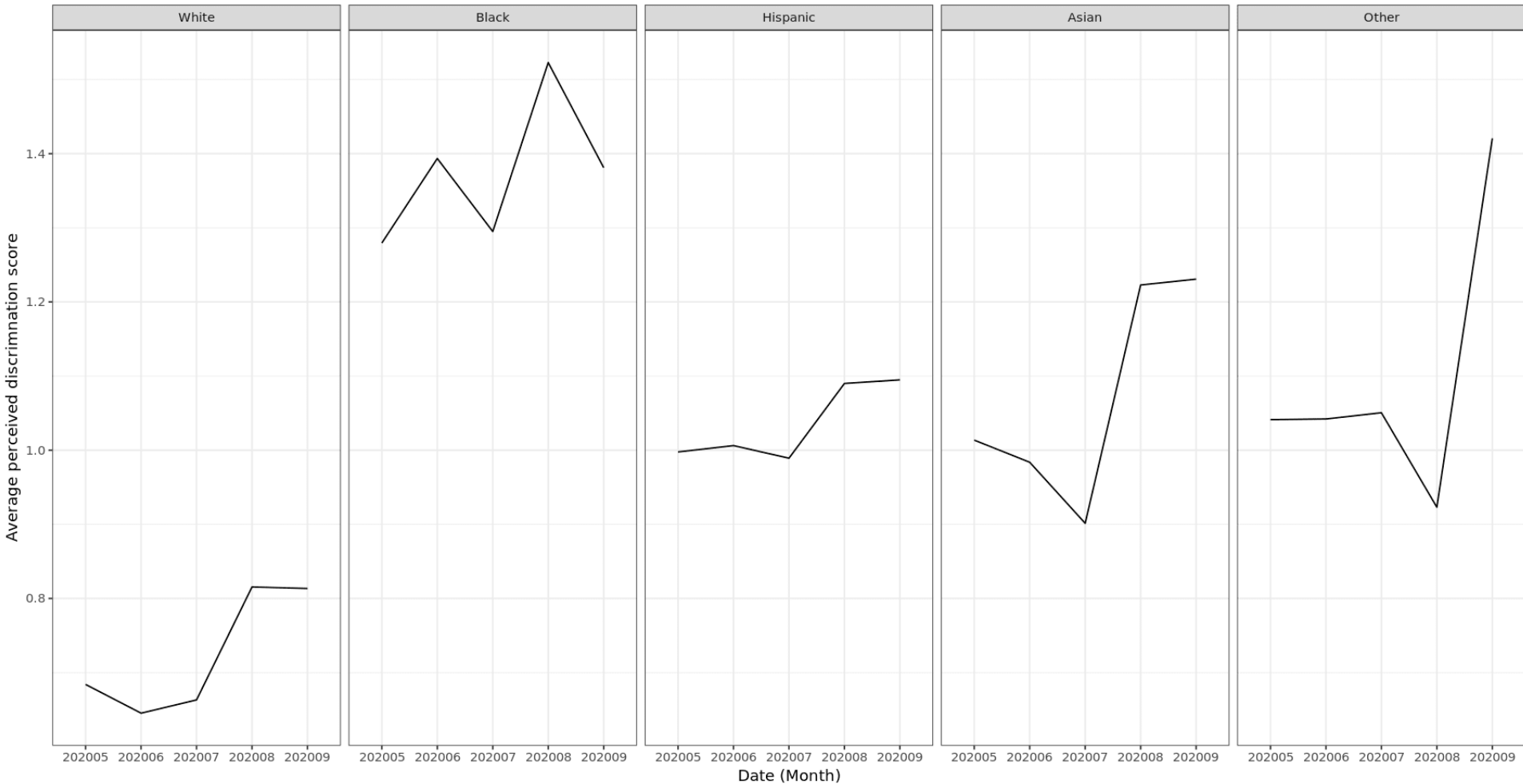
Race breakdown:

Race category	<i>n</i>	Percent
White	53971	81.5%
Black	3715	5.6%
Hispanic	4680	7.1%
Asian	1945	2.9%
Other	1909	2.9%

Distribution of (Major Experience) Discrimination

From monthly survey 2020 May to September

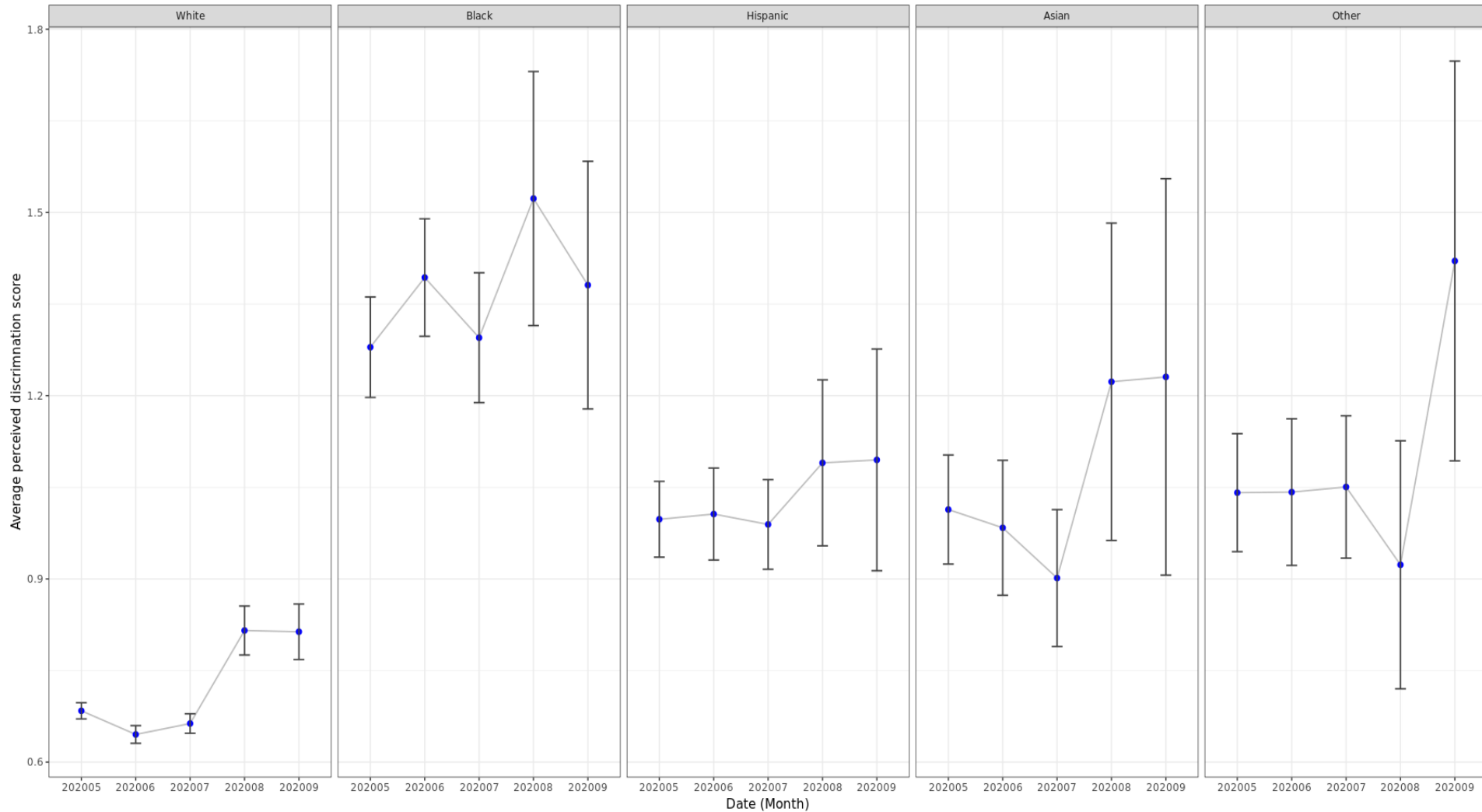
Major Experiences of Discrimination (All of Us, 2020 (N = 67205))



Distribution of (Major Experience) Discrimination

From monthly survey 2020 May to September

Major Experiences of Discrimination



Linear Mixed Model of Race/ethnicity Predicting Trajectories of Major Discrimination Event

```
Linear mixed model fit by REML ['lmerMod']
Formula: score_disc ~ 1 + race_cat + wave0 + (1 + wave0 | race_cat)
Data: dd
```

REML criterion at convergence: -20.9

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.15394	-0.44140	-0.00786	0.52844	2.26650

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
race_cat	(Intercept)	1.039e-02	1.019e-01	
	wave0	1.861e-33	4.314e-17	-1.00
Residual		1.039e-02	1.019e-01	

Number of obs: 25, groups: race_cat, 5

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	0.63014	0.11530	5.465
race_catBlack	0.65008	0.15788	4.118
race_catHispanic	0.31130	0.15788	1.972
race_catAsian	0.34617	0.15788	2.193
race_catOther	0.37124	0.15788	2.351
wave0	0.04707	0.01441	3.266

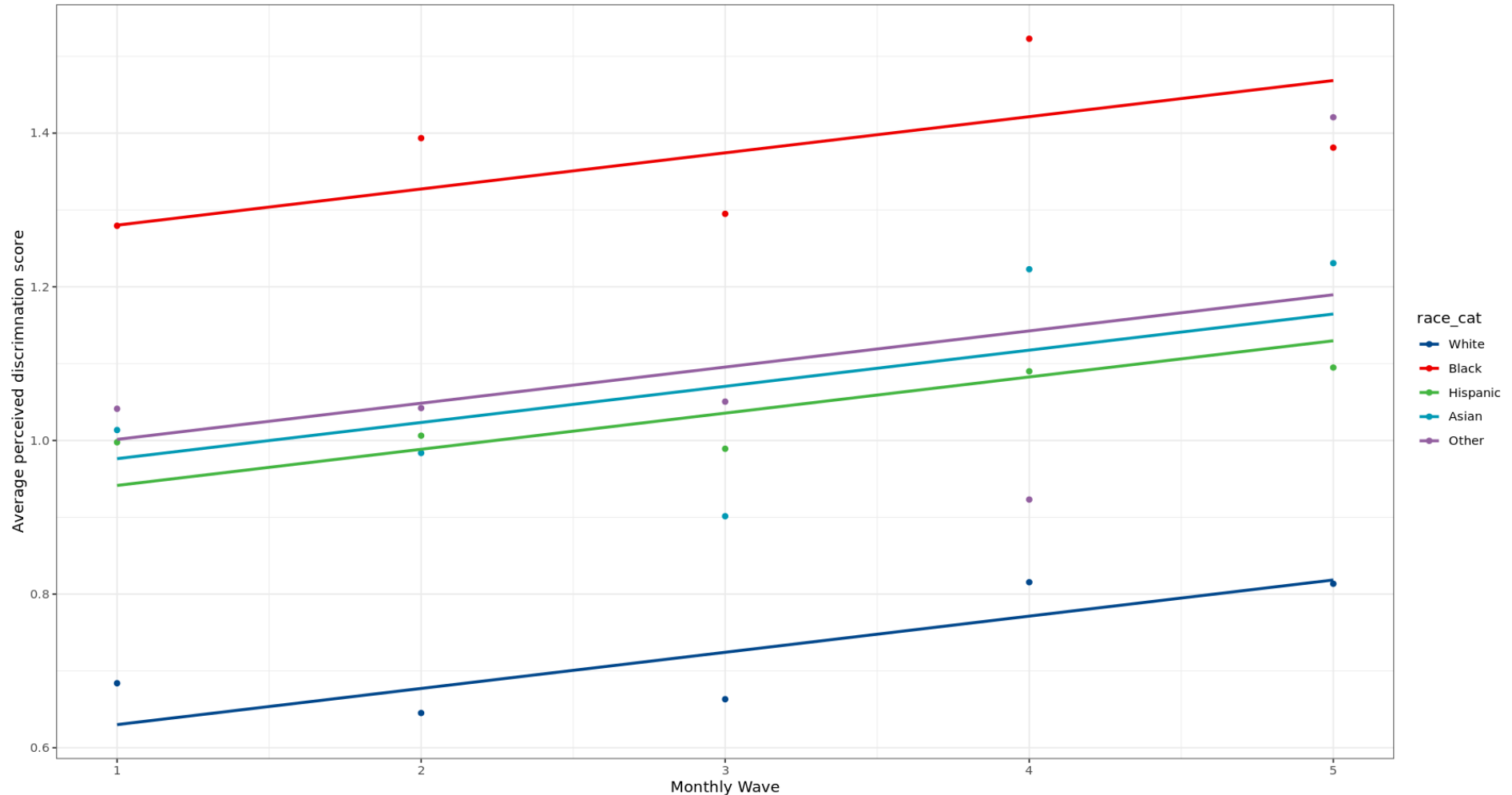
Correlation of Fixed Effects:

	(Intr)	rc_ctB	rc_ctH	rc_ctA	rc_ctO
race_ctBlck	-0.685				
rac_ctHspnc	-0.685	0.500			
race_catAsn	-0.685	0.500	0.500		
race_ctOthr	-0.685	0.500	0.500	0.500	
wave0	-0.250	0.000	0.000	0.000	0.000

This should
be FMLE

Compared to Whites, all of the racial categories have statistically significantly higher rates of major discrimination events during may 2020. In addition, compared to White's all the other racial groups have increasing trajectories from the first data collection point (May 2020) to the last data collection point.

Linear Mixed Model of Race/ethnicity Predicting Trajectories of Major Discrimination Event



Results

- Were my hypotheses supported?
- How do you know? What aspect of the results did you use to come to your conclusion?
- What do you think the meaning of the results may be? What are the implications for the future?

Some Tools to Analyze Longitudinal Data

- Exploratory analysis of correlated data
- Random effect model
- Growth curve models
- Modeling covariance via serial correlation, measurement error
- Generalized Estimating Equations (GEE)
- Analysis of discrete longitudinal data
- Estimation with missing data
- Methods for multivariate longitudinal data
-

Summary

- Longitudinal data can help us understand dynamic changes over time
- We can understand changes in disease incidence and prevalence
- We can understand what predicts dynamic change
- There are rich datasets that we can draw information from
 - Data repositories
(<https://www.icpsr.umich.edu/web/pages/>)
 - AllofUs Research Program

Next Class

- Two key questions related to studying change
- Key features of studying change
- Person-Level (“wide”) data format pros and cons
 - Descriptive statistics
 - Profile plots
- Person-Period (“long”) data format pros and cons
 - Empirical growth plots
 - Non-parametric standardization
 - Parametric standardization
- Functional Form

Class Exercise 1

1. Go to the All of Us Research data browser
 - <https://databrowser.researchallofus.org/>
2. Select the COPE data source
3. Open the PDF of the survey questions
4. Formulate 3 different longitudinal research questions.
 - EX: How does reported self isolation change during the COVID-19 pandemic? Are their differences in this change by age?

Class Exercise 2

1. Go to the Youth Risk Behavioral Survey
 - <https://www.cdc.gov/healthyyouth/data/yrbs/index.htm>
2. Scroll through the website to answer the following questions
 - a. What is the purpose of collecting these data?
 - b. What are the time periods these data are collected?
 - c. Are the data collected on the same person over these collection periods?
 - d. Is this considered longitudinal data?